

Portability of three reported NR4A3 fusion associated RNA signals across extraskkeletal myxoid chondrosarcoma cohorts

Tristan D. McRae

Independent researcher, unaffiliated. Correspondence: trimcrae@gmail.com. ORCID: 0000-0002-1823-1451.

Abstract

Experimental studies have linked NR4A3 fusion proteins to PPARG, SEMA3C and ENO3 expression in different cellular models. Their RNA associations were compared across three public extraskkeletal myxoid chondrosarcoma (EMC) series measured by two array platforms and 3'-end sequencing. ENO3 was elevated on both arrays after multiple-testing correction and remained elevated across comparator strata and a matched hybridization-reference pool. It was a preselected positive control, so its basic elevation is not an independent discovery. PPARG's strongest array result reused the original discovery dataset; its other array result did not survive correction. SEMA3C changed direction between low-grade fibromyxoid sarcoma and desmoid comparators. In the sequencing series, ENO3 had EMC-to-normal and EMC-to-sarcoma ratios of 2.53 and 2.02. These findings distinguish the portability of three previously reported expression associations. Small cohorts, different measurements and incomplete fusion annotations limit generalization, and expression comparisons do not establish fusion causation.

Keywords: Extraskkeletal myxoid chondrosarcoma; NR4A3; ENO3; PPARG; SEMA3C; gene expression

Main text

Experiments have linked NR4A3 fusion proteins to expression of PPARG, SEMA3C and ENO3, but in different cellular models and with different fusion partners. PPARG was investigated with EWSR1::NR4A3, SEMA3C with EWSR1::NR4A3 and TAF15::NR4A3, and ENO3 with TFG::NR4A3.[1–3] These observations motivate a narrower question: how consistently do their RNA associations transfer across public extraskkeletal myxoid chondrosarcoma (EMC) expression series and comparator histologies?

Three series supplied complementary measurements. GSE24369 contained six EMC biopsies and 29 comparison tumors: 17 low-grade fibromyxoid sarcomas, six myxofibrosarcomas and six desmoid fibromatoses.[4] GSE4303 supplied ten EMC tumors and six usable comparators on GPL3290, comprising three dermatofibrosarcoma protuberans and three gastrointestinal stromal tumors.[5] GSE28866 supplied four EMC libraries measured by 3'-end sequencing, 32 other sarcoma libraries representing 30 specimens, and 27 normal-organ libraries.[6] Its duplicate accession GSE170983 was not counted as another cohort. Measurements were analyzed separately because the platforms represent different quantities.

Array expression was standardized against each sample's probe distribution; gene scores averaged readable probes, and contrasts compared EMC with the specified reference group. Exact sample-label permutation tests and within-platform Benjamini–Hochberg correction assessed the retained gene contrasts. Comparator-stratum, reference-pool, leave-one-out, rank-based and tissue-composition sensitivities examined portability. ENO3 had been designated a positive control before

this assessment, preventing its basic elevation from being treated as an independent discovery. Supporting files retain specimen definitions, probe mappings, procedures and all results.

ENO3 was elevated on both array platforms after correction (Table 1). On GPL6244, its differences against low-grade fibromyxoid sarcoma, combined myxoid histologies and desmoid fibromatosis were similar: +0.805, +0.808 and +0.807. Restricting GPL3290 to the three dermatofibrosarcoma protuberans samples sharing the EMC hybridization reference pool retained a difference of +3.515. These comparisons support consistency across the examined references.

PPARG and SEMA3C behaved differently. PPARG survived correction only on GPL3290, whose GSE4303 data underlie the original report of high PPARG expression in EMC.[5] SEMA3C did not survive correction in either pooled comparison. Its direction changed with histology: +1.657 against low-grade fibromyxoid sarcoma and -0.645 against desmoid fibromatosis, with unadjusted stratum P-values of 0.000119 and 0.01515. Against reference-pool-matched GPL3290 comparators, the difference was +0.113 (P=0.843). A pooled contrast therefore conceals substantial comparator dependence.

In the sequencing series, ENO3 had EMC-to-normal and EMC-to-sarcoma ratios of 2.53 and 2.02, ranking at the 98.0th and 95.9th percentiles of genes with computable ratios on those respective axes. SEMA3C ratios were 1.82 and 1.66, and PPARG ratios were 1.42 and 2.12. These descriptive measurements involve few peaks per gene and are not equivalent to array expression. Normal references were visceral organs rather than matched soft tissue. Differentiated skeletal-muscle markers did not reproduce ENO3's tumor contrast, weakening simple muscle admixture as an explanation while leaving tumor-intrinsic differentiation unresolved.

A complementary aggregate combined the three fusion-associated genes with sixteen native-NR4A3 targets. It did not exceed size-matched random-set thresholds on either array, whereas a published EMC signature did.[1] This competitive result does not negate expression differences: the heterogeneous aggregate also showed a GPL3290 group difference under sample-label permutation (P=0.011). Complete aggregate, chromatin-surrogate and adverse results remain in the supporting files.

The contribution is the comparison of three previously reported RNA associations. ENO3 was most consistent across the examined expression checks, PPARG's strongest array result reused discovery data, and SEMA3C depended on comparator histology. Small EMC groups, tissue composition, measurement differences and incomplete fusion annotations limit generalization. These comparisons neither overturn the original experiments in their respective systems nor identify causal fusion targets in EMC tumors. They establish where the expression associations remain consistent across the examined contexts.

Data availability

Public expression series are GSE24369, GSE4303 and GSE28866; GSE170983 duplicates the latter. Extended evidence and its two supporting files retain the methods, all historical results, complete Table S14 and the source/artifact inventories. Selected code, outputs and figures are available at

<https://github.com/trimcrae/Rare-cancers/tree/b8b176913bfa65c711573690b1b3feac7059e393>. The retained 37-file manifest verifies selected artifact identities, not a complete historical environment or independent reproduction. No archival DOI for this fusion-output package has been verified.

Declarations

No external funding supported this work. The author has no competing financial interests and declares a nonfinancial interest as an EMC survivor. The study used public data and reports no new recruitment, specimen collection or experiments.

Tristan D. McRae is the sole author and directed the work. Claude (Anthropic) and OpenAI models assisted with the original literature review, computational work and drafting. OpenAI Codex assisted with this shorter presentation and checks against the retained evidence. The author is responsible for the content; AI assistance does not constitute independent scientific validation.

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Table 1

Array contrasts for the three reported genes. Values are differences in mean standardized expression, EMC minus the stated comparator group. The two platform scales should not be compared numerically. q-values retain Benjamini–Hochberg correction across all six tested genes on GPL6244 and five readable tested genes on GPL3290, including controls. GPL6244 contrasts six EMC with 29 reference tumors. GPL3290 contrasts ten EMC with six reference tumors for ENO3 and SEMA3C; PPARG has five readable reference tumors.

Gene	GPL6244 difference	q-value	GPL3290 difference	q-value
ENO3	+0.8075	0.00044	+3.8113	0.00063
PPARG	+0.3071	0.097	+2.4809	0.00083
SEMA3C	+0.7298	0.233	+0.6228	0.165